

Quiz (Carolina Reaper)

- Q1.** A shipping company has two types of trucks. The first type can carry $3x$ packages and travel $2y$ miles. The second type can carry $2x$ packages and travel $3y$ miles. If the company has 240 packages to deliver and 420 miles of road to cover, what is the value of x ?
- (A) 5
(B) 6
(C) 8
(D) 10
(E) 12
- Q2.** A traffic management system uses x cameras and y sensors to monitor traffic flow. Each camera can monitor $5x$ cars and $2y$ roads, while each sensor can monitor $3x$ cars and $4y$ roads. If the system can monitor 150 cars and 80 roads, what is the value of y ?
- (A) 2
(B) 4
(C) 6
(D) 8
(E) 10
- Q3.** A delivery company uses two types of vehicles: cars and trucks. Cars travel at x miles per hour and trucks travel at y miles per hour. If a car travels 120 miles and a truck travels 180 miles in the same time, what is the ratio of x to y ?
- (A) 1:2
(B) 2:3
(C) 3:4
(D) 4:5
(E) 5:6
- Q4.** A logistics company has x warehouses and y delivery trucks. Each warehouse can store $2x$ packages and each truck can carry $3y$ packages. If the company has 480 packages to store and 720 packages to deliver, what is the value of y ?
- (A) 8
(B) 10
(C) 12
(D) 15
(E) 20
- Q5.** A transportation system uses x buses and y trains to transport people. Each bus can carry $4x$ people and each train can carry $5y$ people. If the system can transport 240 people and the ratio of buses to trains is 3 : 2, what is the value of x ?
- (A) 4
(B) 6
(C) 8
(D) 10
(E) 12
- Q6.** A shipping company has two types of containers: 20-foot and 40-foot. The company has x 20-foot containers and y 40-foot containers. If the company has 120 feet of space and 40 containers to ship, what is the value of x ?
- (A) 10
(B) 15
(C) 20
(D) 25
(E) 30
- Q7.** A traffic flow system uses x traffic lights and y pedestrian crossings. Each traffic light can control $3x$ cars and $2y$ pedestrians, while each pedestrian crossing can control $2x$ cars and $3y$ pedestrians. If the system can control 180 cars and 120 pedestrians, what is the value of y ?
- (A) 4
(B) 6
(C) 8
(D) 10
(E) 12
- Q8.** A delivery company uses x vans and y motorcycles to deliver packages. Each van can carry $5x$ packages and travel $2y$ miles, while each motorcycle can carry $2x$ packages and travel $3y$ miles. If the company has 300 packages to deliver and 420 miles of road to cover, what is the value of y ?
- (A) 6
(B) 8
(C) 10
(D) 12
(E) 15

Open Ended Questions (FRQ)

- Q9.** A logistics company has x warehouses and y delivery trucks. Each warehouse can store $2x$ packages and each truck can carry $3y$ packages. If the company has 600 packages to store and 900 packages to deliver, find the values of x and y .
- Q10.** A transportation system uses x buses and y trains to transport people. Each bus can carry $4x$ people and each train can carry $5y$ people. If the system can transport 360 people and the ratio of buses to trains is 2 : 3, find the values of x and y and interpret the results in context.

Chef's Tips

- Use a graphing calculator to check your work.
- Consider the units of the variables in each problem.
- Think about the real-world implications of your solutions.